

BIG TROUT LAKE, ON, Canada

WMO: 716750

Lat: 53.820N

Lon: 89.900W

Elev: 730

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-29.6	-41.6	0.5	-33.1	-37.3	0.7	-29.2	19.9	-6.3	18.2	-3.4	5.0	290	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	80.4	65.3	77.3	63.6	74.2	62.3	68.3	76.3	66.1	73.8	64.1	71.4	8.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	96.6	71.8	63.1	88.7	69.5	61.2	82.9	68.0	33.1	76.9	31.3	73.7	29.8	71.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.2	15.4	DB	-40.9	87.3	5.4	3.4	-44.7	89.7	-47.9	91.7	-50.9	93.6	-54.8	96.1	(4)
(5)			WB	-40.5	72.6	5.1	3.0	-44.2	74.8	-47.2	76.5	-50.1	78.2	-53.8	80.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	-7.1	-3.7	10.3	27.0	42.2	55.5	61.5	59.0	49.6	36.1	18.0	0.3		
	DBStd	26.55	13.60	13.27	14.68	12.19	9.57	8.64	6.63	7.20	8.40	8.27	12.37	14.63		
	HDD50	8575	1771	1504	1229	691	281	40	3	8	109	438	961	1540		
	HDD65	13145	2236	1924	1694	1140	708	304	150	213	464	896	1411	2005		
	CDD50	995	0	0	0	1	38	204	359	288	98	7	0	0		
	CDD65	91	0	0	0	0	0	19	41	28	3	0	0	0		
	CDH74	717	0	0	0	0	19	189	319	172	18	0	0	0		
	CDH80	109	0	0	0	0	2	31	51	24	1	0	0	0		
	WSAvg	7.5	7.3	7.1	7.2	7.0	7.0	6.9	7.0	7.6	8.2	9.0	8.6	7.5		
	PrecAvg	22.4	0.8	0.6	1.0	1.1	1.9	2.4	3.1	3.3	2.9	1.9	1.6	1.1		
	PrecMax	27.8	1.7	1.3	2.0	2.6	5.1	5.3	5.9	6.1	6.6	3.9	2.4	2.6		
(15) Precipitation	PrecMin	12.0	0.4	0.3	0.2	0.0	0.1	0.6	0.4	0.9	0.5	0.4	0.7	0.2		
	PrecStd	3.9	0.3	0.3	0.4	0.7	1.3	1.1	1.7	1.5	1.8	0.8	0.5	0.6		
	DB	31.1	34.8	49.1	63.7	76.3	83.5	85.3	83.1	76.6	62.7	47.2	33.9			
	MCWB	30.0	33.1	41.4	49.3	57.8	65.5	69.5	68.2	63.7	56.5	43.6	32.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	26.1	30.2	42.1	55.8	69.9	79.1	80.6	78.6	71.0	57.0	40.8	29.1			
	2% MCWB	25.2	28.4	35.8	44.3	54.7	62.4	65.7	66.3	60.9	51.3	37.9	28.1			
	5% DB	21.1	25.2	37.5	50.1	64.5	75.4	77.5	75.0	66.8	52.8	36.8	25.5			
	5% MCWB	20.2	23.5	33.3	40.1	52.3	61.0	64.2	64.1	59.0	48.6	35.0	24.6			
	10% DB	14.6	18.3	33.0	44.8	59.5	71.2	74.3	71.5	63.0	48.5	33.6	21.7			
	10% MCWB	13.9	16.8	30.1	37.3	49.4	59.1	62.8	62.4	57.3	45.1	32.3	21.0			
	0.4% WB	30.2	33.5	42.3	50.6	60.9	69.4	72.7	70.6	66.1	58.1	43.8	32.1			
	0.4% MCDB	31.0	35.0	48.6	61.2	70.9	76.6	80.8	78.9	72.4	62.7	46.7	33.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	25.2	28.6	36.7	44.8	56.7	65.5	68.7	67.8	62.9	52.3	38.4	28.2			
	2% MCDB	26.0	30.3	41.0	54.7	67.2	74.4	76.2	76.0	68.3	55.7	40.2	28.9			
	5% WB	20.2	23.4	33.4	41.0	53.4	62.8	66.3	65.7	60.1	48.9	34.9	24.7			
	5% MCDB	21.1	25.2	37.2	48.9	62.5	71.4	74.2	72.9	65.4	52.6	36.6	25.4			
	10% WB	13.9	16.7	30.2	37.7	50.1	60.3	64.3	63.6	57.3	45.2	32.4	21.0			
	10% MCDB	14.6	18.3	33.4	44.3	58.3	68.6	71.8	69.9	62.3	48.6	33.7	21.7			
	MDBR	17.5	21.9	24.2	23.0	21.5	21.4	20.1	18.3	15.7	10.9	11.6	14.5			
	5% DB	19.2	21.8	23.5	28.5	30.5	25.5	23.5	21.4	21.1	17.6	11.9	14.2			
(36) Mean Daily Temperature Range	MCWBR	18.8	19.9	18.6	18.6	17.7	12.6	11.8	11.6	12.8	12.6	9.9	13.7			
	MCDBR	19.1	21.3	21.9	26.2	27.6	22.2	20.3	19.6	17.9	16.6	11.1	14.6			
	MCWBR	18.8	19.7	17.9	17.7	16.9	13.0	11.7	11.7	12.1	12.5	9.7	14.2			
	MCDBR	18.8	19.7	17.9	17.7	16.9	13.0	11.7	11.7	12.1	12.5	9.7	14.2			
(40) Clear-Sky Solar Irradiance	taub	0.240	0.252	0.269	0.336	0.355	0.355	0.388	0.361	0.331	0.291	0.263	0.225			
	taud	2.236	2.278	2.307	2.304	2.402	2.442	2.382	2.465	2.522	2.576	2.376	2.258			
	Ebn at Noon	236	274	293	283	283	284	271	273	269	260	227	220			
	Edh at Noon	19	25	31	36	35	34	35	31	25	19	16	15			
(44) All-Sky Solar Radiation	RadAvg	327	706	1193	1673	1697	1818	1705	1380	917	508	252	209			
	RadStd	30	43	89	118	148	189	165	120	115	52	32	30			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	-0.94	-1.33	-1.08	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	-0.94	-1.33	-1.08	N/A	N/A	N/A	N/A

Nomenclature: See separate page